If you can't beat 'em, join 'em

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July 15, 2026

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- I'm from the US.
- I've gotten old enough that I regularly make cultural references to before any of you were born.
- I'm a mathematician, not a physicist.


AND YOU MAY ASK YOURSELF
"HOW DID I GET HERE?!"

I ask myself that sometimes.

And yet, every time I thought I was just doing mathematics, physics kept rearing its ugly head.

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And yet, every time I thought I was just doing mathematics, physics kept rearing its ugly head.

Eventually it reached a point where I couldn't ignore it. As they say, if you can't beat 'em, join 'em.

So, often what I'm trying to do is take ideas circulating in physics and bring that insight into mathematics.

I thought that what I was working on ≈20 years ago was **almost commutative algebras**: non-commutative algebras that become commutative in a limit.

In physics terms, you would think about these in terms of a star-product deforming the functions on the phase space of a classical system.

But something special is happening here: since I'm working in an algebraic setting, I can think of these as the holomorphic functions on a complex manifold.

Taking the classical limit gives me a **hyperkähler** structure on this manifold.

So far, so math. But at some point I learned:

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- This structure arises whenever we have a 3d quantum field theory with a topological twist (by considering the local operators in the twisted theory).
- All of the examples I was studying occurred this way!
- If you have a 3d theory with $\mathcal{N} = 4$ supersymmetry, it has two natural classes of topological twists Q_A and Q_B, and thus two hyperkähler manifolds associated to it: the **Coulomb** and **Higgs** branches.

There is a particularly natural 3d gauge theory $\mathcal{T}(G,N)$ for a complex representation N of a complex reductive group G. In this case, the two branches are not created equal:

- $\mathfrak{M}_H$ is classical: the hyperkähler quotient $\mathbb{H}^n////G_c$ (Hitchin–Karlhede–Lindström–Roček, 1987). Mathematicians digested this decades ago.
- $\mathfrak{M}_C$ receives quantum corrections from monopoles. For $\sim$20 years, physicists computed examples of a space with **no mathematical definition**.

Braverman–Finkelberg–Nakajima (2016): a rigorous definition, but quite technical and difficult to understand. I should probably emphasize: this definition was new to physicists!

A number of my research projects in recent years have focused on understanding this definition and its consequences more deeply.

3d mirror symmetry (Intriligator–Seiberg, 1996)

pairs of theories $\mathcal{T}, \mathcal{T}^!$ equivalent in a way that swaps $Q_A \leftrightarrow Q_B$. In particular:

$$\mathfrak{M}_H(\mathcal{T}) \cong \mathfrak{M}_C(\mathcal{T}^!) \qquad \mathfrak{M}_C(\mathcal{T}) \cong \mathfrak{M}_H(\mathcal{T}^!)$$

To a physicist: a duality. To a mathematician: a seemingly inexplicable pairing of spaces with no visible relationship.

Despite the difficulty of understanding this connection, with collaborators, we had guessed a number of examples of dual pairs, and proved some of the consequences of this duality.

Chern–Simons theory:

- Defined by a "sum over histories" (path integral) where the notion of history is a connection on a principal G-bundle over a 3-manifold M.
- Given a knot $K \subset M$ and a representation V of G, we can define the Wilson loop observable $W_{K,V}$, which is an expected value of the trace of the holonomy of the connection around K in the representation V.

Witten (1989): expectation values of Wilson loops in Chern–Simons theory are polynomials in a parameter q; $\mathrm{SU}(2)$ with the fundamental representation gives the Jones polynomial.

Question

How does the invariant for $\bigwedge^m \mathbb{C}^N$ over $\mathrm{SU}(N)$ vary as a function of m, N, q?

There is expected to be a recursion in m and N (the **quantum $\hat{A}$-polynomial**).

Physics:

- should be particularly simple in the **planar ('t Hooft) limit**:

$$N \to \infty, \qquad q \to 1, \qquad g = q^N \text{ fixed}$$

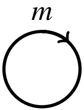
- should have a holographic dual description as a topological string on the resolved conifold, with a brane on the conormal Lagrangian of the knot.

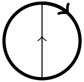
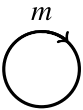
This dual story has also been studied mathematically, and has its own invariants: the **augmentation variety** of the knot, defined by counting holomorphic curves with boundary on the brane. This is a Lagrangian subvariety of $(\mathbb{C}^*)^2$ (or $(\mathbb{C}^*)^3$ if we include g).

Thus we have two independently defined invariants of a knot: the augmentation variety, and the planar limit of the quantum $\hat{A}$-polynomial. Aganagić–Vafa (2012) predicted that these are equal.

Idea (after Gaiotto–Kannagi–Sanjurjo): treat the coloring m as a **formal variable**. While this does not make sense as a representation of $\mathrm{SU}(N)$, we can still make skein theoretic calculations.

For the unknot, substituting $g = q^N$, $\mu = q^m$ (no N, no m!):



$= \frac{q-q^{-1}}{\mu-\mu}\ m-1$

$m \quad = \frac{qg\mu^{-1}-q^{-1}g^{-1}\mu}{\mu-\mu^{-1}}\lambda$


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So $\mu - \mu^{-1} - \lambda\left(qg\mu^{-1} - q^{-1}g^{-1}\mu\right)$ annihilates the unknot: its deformed $\hat{A}$-polynomial, with no computation of any knot invariant.

Theorem (W.–Zaimi, 2026)

At $q = 1$, the HOMFLYPT difference module **is** the degree-0 abelianized knot contact homology of K.

- The proof is purely algebraic: diagram calculus. No holomorphic curves, no path integrals. Elementary tools, non-elementary conclusion.
- First explicit mathematical connection between these two worlds.

This paper owes a large debt to conversations with Davide Gaiotto—the kind of accident that only happens in this building.

Just two examples of these sorts of connections:

- Physics predicted equalities between independently defined objects: Higgs $\leftrightarrow$ Coulomb; recursions $\leftrightarrow$ augmentation varieties.
- Proving them required inventing structures that now have lives of their own.

Physics and mathematics have a few centuries of profitable interaction behind them, and yet we keep finding new ways to surprise each other.